ADAM TANG

Embedded hardware engineer and creative ideator with end-to-end product development experience, from circuit design, mechatronics and prototyping to system integration and manufacturing

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EDUCATION

University of California, Berkeley — Berkeley, CA
Bachelor of Science in Electrical Engineering and Computer Sciences, Minor in Bioengineering — *Expected June 2028*
- **Relevant Coursework**: Signals + Dynamical Systems Processing | Data Structures | Linear Algebra | Multivariable Calculus

Phillips Exeter Academy | High School Diploma — Exeter, NH
- Awards/Distinctions: 1600 SAT, F1 in Schools 1st Place State Finals 2023 (Mechanical Lead) — *September 2021 - June 2025*

WORK EXPERIENCE

Eldaeon (Skydeck Startup) — Berkeley, CA
Embedded Systems Intern — *December 2025 - Present*
- Designed autonomous antenna pan–tilt protocol w/ geometric targeting ±10° sweep optimization; performed RF system-level validation to maximize wireless communication reliability
- Built multi-sensor system integration pipeline (GPS, compass, RF) ranking top 10 transmitters and selecting optimal target
- Designed DPOL power management architecture, reducing power consumption by **30.2%** via embedded firmware
- Built radar track alert system delivering real-time notifications to user phone and email using AWS SES and Twilio

CalSol (UC Berkeley Solar Racing Team) — Berkeley, CA
Solar Engineer — *September 2025 - Present*
- Optimized solar array layout and bypass diode configuration for Gen 11 vehicle; improved photovoltaic efficiency by **12%**
- Integrated wiring and lightening holes into carbon fiber composite shell, applying design for assembly (DFA) principles

BEACN (Bay-Area Environmentally Aware Consulting Network) — Berkeley, CA
Systems Modeling Engineer (Associate Consultant) — *February 2026 - Present*
- Engineered energy optimization model and GIS for AI data center control company Vigilent, using statistical analysis to quantify optimal efficiency targets by weighing 5–10% efficiency gains, $/kWh, and Power Usage Efficiency (PUE)

MUREX Exeter (MATE Marine Robotics Competition Team) — Exeter, NH
Mechanical Lead — *November 2021 - June 2024*
- Led design and fabrication of custom 6-DOF underwater ROV, 7-DOF manipulator arm, + buoyancy module; applied design for manufacturability (DFM) to cut manufacturing costs by **88%** through in-house component design
- Managed prototyping and validation across multi-functional teams; collectively broke **10** industry records (incl. world's smallest network switch) and placed 6th globally at MATE 2024

TECHNICAL PROJECTS / RESEARCH

PolySynth X - Expressive MPE Keyboard & Embedded Synth Platform (in progress) | *November 2025 -* — Berkeley, CA
- Led R&D lifecycle from hall sensor circuit analysis to embedded firmware optimization and system-level testing
- Developed and optimized compliant key mechanism for per-key pitch modulation, DSP engine w/ Daisy seed microcontroller

Low-Cost Modular Biodiversity Sensing Module (in progress) | *September 2025 -* — Berkeley, CA
- Developing solar-powered wildlife sensing nodes with camera, acoustic, and thermal inputs; lead on embedded hardware
- Utilizing WiFi telemetry for communication with edge AI for optimized species identification in remote field locations

Microplastic Pollution in Rivers (Catholic University of America) | *June 2022 - August 2024* — Washington DC
- Simulated vertical transport of microplastics using flume hydraulics; generated 250+ data points through dispersion modeling
- Developed drone sampling protocol to access 3+ previously unreachable zones, enabling full environmental diagnostics
- **First-author** research paper selected for WaterSciCon 2024; project awarded by NASA, EPA, GENIUS Olympiad, and ISEF

Fully Automated Aquaponics System | *December 2023 - June 2025* — Exeter, NH
- Designed and built closed-loop aquaponics system with **automated pH, biofiltration, ammonia, and temperature sensing**
- Acquired iterative feedback from national/international experts (UNH, Victory Aquaponics, Island School (Bahamas)); led cross-functional ops with faculty/staff to deploy system as live food + edtech solution

LEADERSHIP

Innovation Towards the Future (NGO) — Potomac, MD
Founder & President — *September 2020 - June 2025*
- Represented the **USA** as youth delegate at the 28th UN Climate Change Conference (COP28) in Dubai, chosen from **5k applicant NGOs**; participated in formal global climate negotiations and policy formulation; NGO admitted to UNFCCC
- Led NGO to forge **3 corporate partnerships** to distribute 1000+ eco-friendly bags/cups to communities in need
- Launched and promoted global No Single-Use Plastic Campaign through national and international network connections

SKILLS

Programming: Python, Java, SQL, C/C++, MATLAB, TypeScript, JavaScript, AI/ML + Pytorch (learning)
Hardware Design: Circuit Analysis, Digital Logic Design, Schematic Capture, PCB Design, RF Circuit Design, Power Management
Engineering Tools: ANSYS (FEA), Solidworks, Fusion 360, Onshape, 3D Printing, Microcontrollers, Oscilloscopes
Systems & Manufacturing: System Integration, Design Verification, Embedded Systems/Firmware, DFM, DFA, Rapid Prototyping